TED(15)-5043 Register Number-----------
(REVISION-2015) Signature ------------

DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY/MANAGEMENT/COMMERCIAL PRACTICE

Control Systems

[Time : 3 Hours]

(Maximum marks : 100)

PART – A

(Maximum marks : 10)

I. Answer the following questions in one or two sentences. Each question carries 2 marks

1. What is centroid of a root locus?.
2. Define transfer function of a control system.
3. Define marginal stability.
4. Define the transient and steady state response of a control system.
5. Write the Laplace transform of a unit ramp function.

(5 X 2 = 10)

PART – B

(Maximum marks : 30)

II. Answer any five of the following questions. Each question carries 6 marks

1. Derive Laplace transform of the function e^{at}.
2. Define transfer function. Explain the term order of a transfer function
3. Briefly explain about linear time variant and linear time invariant systems.
4. Derive the response of a first order system to a unit step input.
5. Describe Force-Voltage and Force -Current analogy.
6. Construct Routh array and determine the stability of the system represented by the characteristic equation $s^5+s^4+2s^3+2s^2+3s+5=0$. Comment on the location of roots of the characteristic equation.
7. Find the magnitude and phase of the integral factor $\frac{k}{s}$ and draw its Bode plot..

(5 X 6 = 30)

PART – C

(Maximum marks : 60)

(Answer one full question from each unit. Each question carries 15 marks)

Unit I

III.

a) Define Find the Laplace Transform of the differential equation given below and hence evaluate the time solution of the same given that $y(0^+)=0$ and $y'(0^+)=6$.

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 12e^t. \qquad (10)$$

b) Explain the mathematical model of a control system. (5)

OR

IV.

(a) With the help of a neat block diagram explain open loop system and closed loop system. (10)

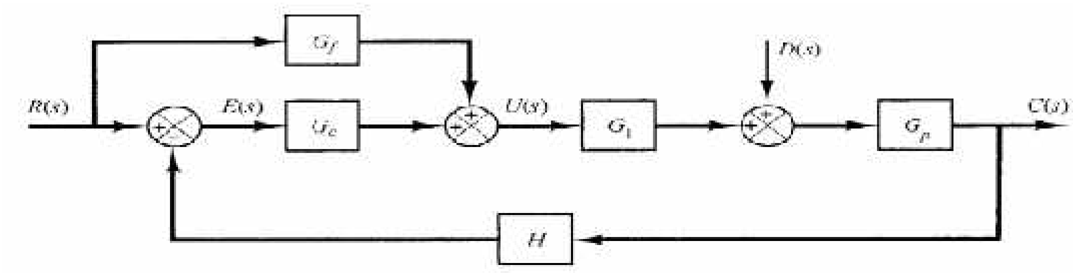
(b) Determine the inverse Laplace transform of the function X(s)= s/(s+3)(s+6). (5)

Unit II

V.

(a)Derive the general transfer function of a Mechanical Rotational system. (10)

(b)For the block diagram shown draw the signal flow graph. (5)



OR

VI.

(a) Draw the circuit diagram of a parallel RLC circuit and derive its transfer function. (8)

(b) Find the response of a first order system for unit step input. (7)

Unit III

VII.

(a) For a unity feedback control system the open loop transfer function is $G(s) = \frac{10(s+2)}{s^2(s+1)}$.

find (i) the position, velocity and acceleration error constants. (ii) the steady state error when the input $R(s) = \frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}$ (9)

(b)Determine the stability of the system using Routh Hurwitz criteria. (6)

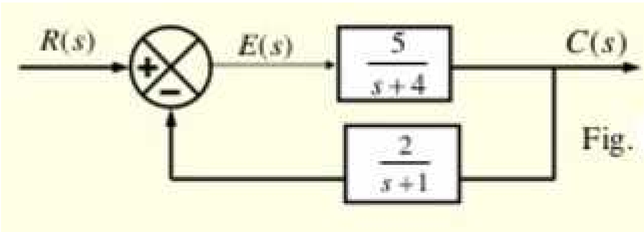
$$s^5+3s^4+2s^3+6s^2+6s+9=0$$

OR

VIII.

(a) With necessary steps draw the Bode plot for the function $1+sT$. (9)

(b) Find the steady state error for the system shown when the input r(t) is (i)unit impulse (ii)unit step and (iii)unit ramp. (6)



Unit IV

IX.

a) Define the terms :

i. Gain cross over frequency
ii. Phase cross over frequency
iii. Phase margin
iv. Gain margin (8)

(b) Draw bode plot for the open loop transfer function G(s)H(s)= s+5 (7)

OR

X.

(a) Sketch the root locus of the unity feedback system whose open loop transfer function is given as $G(s) = \frac{10}{s+2}$ (8)

(b) Draw bode plot for the open loop transfer function G(s) H(s) = K for K=1, K>1 and K<1 (7)